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EXPERIMENT 14
Pro code extension with Catalyst

Aim

Extend the app with a small authenticated Catalyst service and compare it with a Creator custom function.

Procedure

1. Choose a bounded extension: validate the Enquiry ID format (e.g. EQ005) and return a normalized value. It must not authorize admission or expose student data.
2. Create a Catalyst Advanced I/O function (Node.js) with POST /validate-enquiry-id and secure it with Catalyst authentication or API Gateway rules.
3. Call it from Creator through a secured connection from a staff-only action; do not call on every keystroke.
4. Handle timeouts and bad JSON separately from 'invalid format'.

```
function validateEnquiryId(v){
  if(typeof v!=="string") return {valid:false,reason:"id must be text"};
  const r=v.replace(/[ -]/g,"").toUpperCase();
  return {id:r, valid:/^EQ\d{3,6}$/.test(r)};
}
```

Adjust the pattern if your institution uses another enquiry numbering scheme.

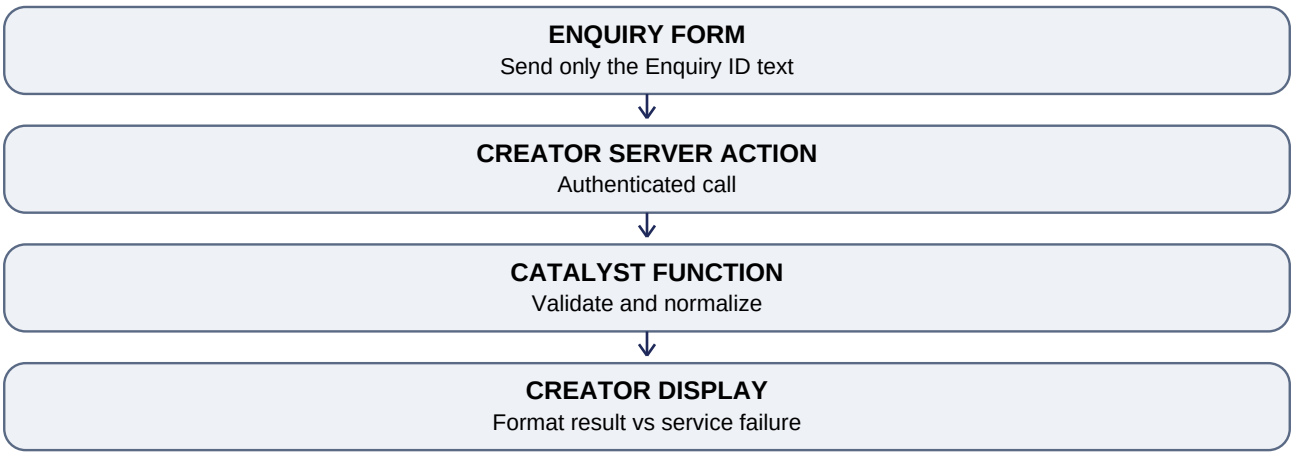
Implementation and verification

Input	Expected
eq005	Normalized EQ005, valid
EQ 005	Normalized EQ005, valid
5	Invalid format
Null or number	Rejected (400)
No credentials	401 or 403

Evidence and testing

E14: function code, interface contract, authenticated Creator call, test results. This is a format check only, not proof the enquiry exists. Interface contract: request {"id": "eq005"}; response {"id": "EQ005", "valid": true}. Actual test results: _____.

Process diagram



Generated explanatory diagram. It is not a screenshot of an executed workflow.

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Result

Expected result and verification criteria. Valid and invalid inputs are distinguished from service failures; you can justify when an external service is worth using versus a local Creator function.